The economy of *Yhtälösade*:
when can a resource-limited equation game never be lost?

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Abstract

In *Yhtälösade* linear equations fall down a well and the player solves them by dragging operation bricks ($+\mathsf{x}$, $-\mathsf{x}$, $+\mathsf{1}$, $-\mathsf{1}$) onto them; every use costs one unit of a finite stock. A solved equation $x = \xi$ becomes a bomb: for $\xi > 0$ it returns ξ units to the stock, split at random between the bricks, and for $\xi < 0$ it clears trash. We ask which constraints on the solutions $\xi_1, \ldots, \xi_N$ and on the operations needed make the game impossible to lose for a player who solves optimally. We show that for the current level the cheapest solution is unique, so the game reduces to an exact ledger of demand and reward vectors. A conservation law then yields a balance inequality that any never-losing design must satisfy. The current parameters violate it by $457/404 \approx 1.13$ units per equation, and even a clairvoyant reward rule could not save more than 16% of 20-equation games. The random split makes things worse: we prove that under it no sure guarantee exists short of a starting stock covering all future demand. We then give two designs with sure guarantees, both with proofs: a *deficit-filling* reward rule, whose condition is a Lindley-queue inequality on the sequence $(\delta_i - \xi_i{}^+)$, and an *adaptive generator* that works with any reward rule. Finally we extend the cost theory to doubling, halving and square roots. The cost of an equation then becomes a capped signed-binary weight, the order of the operations starts to matter, and $\pm$-bombs act as automatic stabilisers of the economy.

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1 Introduction

Yhtälösade ("equation rain") is a Tetris-like game for Finnish upper-secondary students. One equation at a time falls from the top of a well with brick walls. Four special bricks in the walls carry the operations $+\mathsf{x}$, $-\mathsf{x}$, $+1$ and -1, each with a count. Dragging a brick onto the falling equation applies its operation to both sides and lowers the count by one. When the equation reads $x = \xi$ it turns into a bomb labelled ξ:

- a positive bomb returns ξ units of operations, split at random between the bricks;
- a negative bomb clears trash around the place where it lands.

An equation that reaches the ground unsolved stays there as trash, and trash above a red line ends the game.

In the first test game the designer, who believes they played optimally, ran out of $-\mathsf{x}$ at the third equation and out of -1 at the fifth. After that only $+1$ was left, and almost nothing could be solved. This note explains why that outcome is typical, not bad luck. It then answers the designer's question: *what constraints on the solutions $\xi_1, \dots, \xi_N$ and on the operations they require guarantee that an optimally playing student never loses?*

Results in brief.

(i) On the current level the optimal play for each equation is unique up to order and uses a demand vector $d(E)$ that does not depend on the solution (Proposition 3.3). The game therefore reduces to an exact ledger (Section 4).

(ii) *Balance law* (Theorem 4.2). A perfect player survives N rounds only if, for every i, $\sum_{t\le i} \delta_t - \sum_{t<i} {\xi_t}^+ \le \|c_0\|$, where δ_t is the number of operations equation t needs and $\|c_0\|$ the starting stock. In the long run this means $(1-q)\,\nu_+ \ge \bar{\delta}$, where q is the share of negative solutions and ν_+ the mean positive solution. The current generator has $(1-q)\nu_+ = 1.75$ against $\bar{\delta} = 2.88$.

(iii) *Random rewards cannot be relied on* (Theorem 5.3). Under the current random split, survival is certain only if the starting stock covers the total future demand, brick by brick. A perfect player solves all 20 equations in only about 0.2% of games, and the first stall comes on average at equation 5.5 (Table 2).

(iv) *Sure guarantees.* With a deficit-filling reward rule, survival is guaranteed whenever a Lindley queue driven by $(\delta_i - {\xi_i}^+)$ stays below the starting buffer (Theorem 6.3). One concrete instance: every negative solution is followed by a positive one that repays it. With $c_0 = (5,5,5,5)$ this keeps half of the solutions negative and all of them in $\pm\{1,\dots,6\}$ (Corollary 6.4). Separately, a generator that looks at the stock guarantees survival under *any* reward rule (Theorem 6.8).

(v) *Extensions* (Sections 8 to 10). With $\times 2$ or $\div 2$ the cheapest cost is $m + w_m(A) + w_m(B)$, with w_m a capped signed-binary weight. The order of the operations matters, and the cheapest demand vector need not be unique. Square roots create ±-bombs that both refuel and clean: a third of symmetric quadratics is enough to cancel the current deficit.

All numerical claims were checked by exact computation, by exhaustive search or by Monte Carlo simulation; see Section A.

2 The model

2.1 Equations and operations

Throughout, an *equation* is $U(x) = L(x)$ with affine sides and

$$U(x) - L(x) = \Delta\,(x - \xi), \qquad \Delta > 0,\ \xi \in \mathbb{Z} \setminus \{0\}. \tag{1}$$

U is the side with the larger coefficient of x and ξ is the solution. (To avoid a clash with the variable x, we write ξ_i for the solutions $x_1, \ldots, x_N$ of the question.) On the current level $\Delta = 1$. The *bricks* form a finite set J, on the current level $J = \{+\mathsf{x}, -\mathsf{x}, +1, -1\}$, and brick j applies its operation to both sides. A *stock* (inventory) is a vector $c \in \mathbb{N}^J$, and its *total* is $\|c\| := \sum_j c(j)$, the number of operations it holds. This is the ℓ^1 norm, which on $\mathbb{N}^J$ is the plain sum; we use it for all vectors in $\mathbb{N}^J$ (stocks, demands, rewards, counts of operations used), while $|z|$ is always the absolute value of an integer. Inequalities between vectors are componentwise, and $z^+ = \max(z, 0)$, $z^- = \max(-z, 0)$.

2.2 A round of the game

Equations $E_1, E_2, \ldots$ fall one at a time; the next one appears only after the previous one has been resolved. In round i, with stock c_{i-1}:

- each operation the player applies costs one unit of its brick, and only a brick with a positive count can be used;
- as soon as the equation reads $x = \xi$ or $\xi = x$ it locks and becomes a bomb;
- a bomb with $\xi > 0$ adds a *reward* $r_i \in \mathbb{N}^J$ with $\|r_i\| = \xi$ to the stock, chosen by an *allocation rule*;
- a bomb with $\xi < 0$ adds nothing but clears trash (Section 7);
- an unsolved equation that lands becomes a trash block one row high.

The current level ("D0"). The parameters are:

- starting stock $c_0 = (3, 3, 5, 5)$ (order $+\mathsf{x}, -\mathsf{x}, +1, -1$) and $N = 20$ rounds;
- ξ uniform on $\pm\{1, \ldots, 6\}$;
- each equation is built backwards from $x = \xi$ by adding k_b boxes and a constant k to both sides: $(1 + k_b)x + k = k_b x + \xi + k$, with the sides swapped at random;
- k_b uniform on the multiset $\{-2, -1, -1, 0, 1, 1, 2\}$, k uniform on $\{-3, \ldots, 3\}$; pairs with $|k_b| + |k| \notin [1, 4]$ are rejected, and those with $|k_b| + |k| = 1$ are kept with probability $2/5$;
- the random split rule: $k \sim U\{1, \ldots, \min(\xi, 4)\}$ parts, each starting at 1, the remaining $\xi - k$ units added to uniformly random parts, and the parts sent to k distinct random bricks.

Definition 2.1 (Perfect player, stall, survival)**.** The *perfect player* has no time pressure, wastes no operations, and solves the current equation whenever some solving sequence is affordable, using a cheapest affordable one. Round i *stalls* if no solving sequence is affordable with c_{i-1}. The game is *strongly survived* over N rounds if no round stalls.

"Never lost with optimal bombing" is taken to mean strong survival of the perfect player. Section 7 shows that the game's formal end, Game Over, is out of reach in a 20-equation game on a typical phone. The loss players actually experience is the stall: once the stock is depleted, every later equation turns into trash.

3 Solving cost on the current level

Lemma 3.1 (Invariants)**.** *Every operation preserves $U - L$, hence Δ and ξ, and U stays the side with the larger coefficient.*

Table 1: Exact demand statistics of the generators. D0 is the current level; D1 is the rebalanced proposal of Section 11 ($|k_b| \le 2$, $|k| \le 2$, $\delta \le 3$, cost-1 pairs kept with probability 1/2, $\xi \in \{-6, \dots, -1\} \cup \{3, \dots, 8\}$).

	$\mathbb{P}(\delta{=}1,2,3,4)$	$\mathbb{E}\delta$	$\mathbb{E}d(\pm\mathsf{x})$	$\mathbb{E}d(\pm 1)$	$\mathbb{E}\xi^+$	$\mathbb{E}[\xi^+ - \delta]$
D0	$\frac{6}{101}, \frac{30}{101}, \frac{35}{101}, \frac{30}{101}$	$\frac{291}{101} \approx 2.881$	$\frac{57}{101} \approx 0.564$	$\frac{177}{202} \approx 0.876$	$\frac{7}{4}$	$-\frac{457}{404} \approx -1.131$
D1	$\frac{1}{9}, \frac{4}{9}, \frac{4}{9}, 0$	$\frac{7}{3} \approx 2.333$	$\frac{5}{9} \approx 0.556$	$\frac{11}{18} \approx 0.611$	$\frac{11}{4}$	$+\frac{5}{12} \approx +0.417$

Proof. Each of $+\mathsf{x}, -\mathsf{x}, +1, -1$ adds the same affine function $g \in \{x, -x, 1, -1\}$ to both sides, so $(U + g) - (L + g) = U - L$. □

By (1) with $\Delta = 1$, every level-one equation with solution ξ can be written uniquely as

$$U = (1 + p)x + q, \qquad L = px + (\xi + q), \qquad (p, q) \in \mathbb{Z}^2. \tag{2}$$

The generator produces $(p, q) = (k_b, k)$.

Lemma 3.2 (Coordinates)**.** *The operations act on (p, q) as unit translations: $+\mathsf{x}\colon p \mapsto p + 1$, $-\mathsf{x}\colon p \mapsto p - 1$, $+1\colon q \mapsto q + 1$ and $-1\colon q \mapsto q - 1$. The equation reads $x = \xi$ or $\xi = x$ if and only if $(p, q) = (0, 0)$.*

Proof. The first claim is immediate from (2). For the second, a solved equation has one side equal to x and the other a constant. By Lemma 3.1 the coefficient of U exceeds that of L by one, so $U = x$ and $L = \xi$, i.e. $1 + p = 1$ and $q = 0$. Conversely $(p, q) = (0, 0)$ gives $U = x$, $L = \xi$. □

Proposition 3.3 (Cost and forced demand)**.** *Let n_j be the number of uses of brick j in a sequence of operations. The sequence solves $E = (p, q; \xi)$ if and only if $n_{-\mathsf{x}} - n_{+\mathsf{x}} = p$ and $n_{-1} - n_{+1} = q$. Consequently*

$$n \;\ge\; d(E) := (p^-,\; p^+,\; q^-,\; q^+)$$

componentwise for every solving sequence. The cheapest cost is $\delta(E) := \|d(E)\| = |p| + |q|$, and it is attained exactly by the sequences that use $d(E)$, in any order.

Proof. The translations commute, so the final point is $(p - n_{-\mathsf{x}} + n_{+\mathsf{x}},\, q - n_{-1} + n_{+1})$, and Lemma 3.2 gives the first claim. From $n_{-\mathsf{x}} = p + n_{+\mathsf{x}}$ we get $n_{-\mathsf{x}} \ge p$ and $n_{-\mathsf{x}} \ge 0$, so $n_{-\mathsf{x}} \ge p^+$; likewise $n_{+\mathsf{x}} \ge p^-$, $n_{-1} \ge q^+$ and $n_{+1} \ge q^-$. Summing gives $\|n\| \ge |p| + |q|$, with equality if and only if every inequality is tight, i.e. $n = d(E)$. A minimal sequence never passes through $(0, 0)$ before its end, so the lock-on-solve rule does not interfere. □

Corollary 3.4 (Optimal bombing is unique and dominant)**.** *On the current level the perfect player solves E with exactly $d(E)$. Among all ways of solving E this leaves the largest stock in every brick. Hence the whole economy is described by the pairs (d_i, ξ_i), $d_i := d(E_i)$, $\delta_i := \|d_i\|$.*

Remark 3.5 (Negative solutions cost the same). The demand $d(E)$ does not depend on ξ. The reflection $(p, q, \xi) \mapsto (p, -q, -\xi)$, which turns $ax + b = cx + d$ into $ax - b = cx - d$, preserves δ and only swaps the roles of $+1$ and -1.

4 The ledger and the balance law

By Corollary 3.4, as long as no round stalls the stock evolves as

$$c_i = c_{i-1} - d_i + r_i, \qquad \|r_i\| = {\xi_i}^+, \tag{3}$$

and round i stalls if and only if $c_{i-1} \not\geq d_i$.

Proposition 4.1 (Conservation)**.** *If rounds* $1, \dots, i$ *are all solved by the perfect player, then* $\|c_i\| = \|c_0\| + \sum_{t \le i}({\xi_t}^+ - \delta_t)$*. A real player who wastes* $W_i \ge 0$ *operations ends with* W_i *units less.*

Proof. Take the norm of (3), using $\|d_t\| = \delta_t$ and $\|r_t\| = {\xi_t}^+$. □

Theorem 4.2 (Balance law)**.**

(a) If the perfect player strongly survives rounds $1, \dots, N$*, then for every* $i \le N$

$$\sum_{t \le i} \delta_t - \sum_{t < i} {\xi_t}^+ \;\le\; \|c_0\|. \tag{4}$$

(b) If an infinite sequence is strongly survived, then $\liminf_N \frac{1}{N} \sum_{t \le N}({\xi_t}^+ - \delta_t) \ge 0$*. In terms of the empirical share* q *of negative solutions, the mean positive solution* ν_+ *and the mean demand* $\bar{\delta}$*, this is the long-run condition*

$$(1 - q)\, \nu_+ \;\ge\; \bar{\delta}. \tag{5}$$

If in addition the stock stays bounded, (5) *holds with equality in the limit.*

Proof. (a) Round i is solvable only if $c_{i-1} \ge d_i$, which forces $\|c_{i-1}\| \ge \delta_i$; insert Proposition 4.1. (b) Dividing (4) by N gives $\frac{1}{N} \sum_{t<N}({\xi_t}^+ - \delta_t) \ge (\delta_N - \|c_0\|)/N \ge -\|c_0\|/N$. Moreover $\frac{1}{N} \sum {\xi_t}^+ = (1-q)\nu_+$ and $\frac{1}{N} \sum \delta_t = \bar{\delta}$. A bounded stock also bounds the sum from above by Proposition 4.1. □

The balance law is a property of the sequence alone, so no reward rule can repair a violation of it. For the current level, Table 1 gives $\mathbb{E}[\xi^+ - \delta] = -457/404$. By the strong law of large numbers, every long game therefore stalls almost surely. For $N = 20$ the bound is quantitative:

Theorem 4.3 (Allocation-free upper bound)**.** *Let the rounds be i.i.d. Under* any *allocation rule, even one that knows the future,*

$$\mathbb{P}(\textit{strong survival over } N \textit{ rounds}) \;\le\; \bar{p}_N := \mathbb{P}\Big(\forall i \le N :\; \|c_0\| + \sum_{t<i}({\xi_t}^+ - \delta_t) \ge \delta_i\Big).$$

For D0 with $\|c_0\| = 16$*, exact dynamic programming over the total stock gives* $\bar{p}_5 = 0.987$*,* $\bar{p}_{10} = 0.634$ *and* $\bar{p}_{20} = 0.162$*.*

Proof. By Theorem 4.2(a), strong survival implies the event in $\bar{p}_N$, and that event depends only on (δ_t, ξ_t), not on how rewards are split. □

So with the current generator at most 16% of 20-equation games could be survived even by an ideal reward rule; the random split achieves 0.17% (Table 2). The gap is due to misallocation, which is the subject of Section 5. Once the stock is low, the game cannot recover by itself:

Proposition 4.4 (Poverty trap)**.** *The empty stock is absorbing, since every equation has* $\delta \ge 1$*. Under D0, a stock holding only* $+1$ *(at least three units) solves the next equation with probability* $6/101 \approx 0.059$*, and earns a reward with probability* $3/101 \approx 0.030$*.*

Proof. With only $+1$ available, $d(E) \le c$ means $p = 0$ and $-3 \le q < 0$. From the generator's joint law (Section A) these pairs have total probability $\frac{1}{101} + \frac{5}{202} + \frac{5}{202} = \frac{6}{101}$, and $\xi > 0$ independently with probability $\frac{1}{2}$. □

This is exactly the designer's end state: about 34 rounds per rewarded solve on average.

5 Random allocation

Definition 5.1. An allocation rule is *η-diffuse* ($\eta > 0$) if, for every positive bomb and every brick j, $\mathbb{P}(r(j) = 0 \mid \text{past}) \geq \eta$.

Lemma 5.2. *The current split rule, implemented with an unbiased shuffle, is $\frac{3}{8}$-diffuse and gives $\mathbb{E}[r(j) \mid \xi] = \xi/4$ for every j.*

Proof. With $M = \min(\xi, 4)$ and k parts on k random bricks out of four, $\mathbb{P}(r(j) = 0) = \frac{1}{M}\sum_{k=1}^{M}(1 - \frac{k}{4}) = 1 - \frac{M+1}{8} \geq \frac{3}{8}$. The bricks are exchangeable, so each brick receives $\xi/4$ on average. □

Theorem 5.3 (Random rewards cannot be relied on)**.** *Fix a sequence $E_1, \ldots, E_N$ and an η-diffuse rule. Strong survival has probability one if and only if $c_0 \geq \sum_{t \leq N} d_t$.*

Proof. ($\Leftarrow$) Rewards are non-negative, so as long as no round has stalled, $c_{i-1} \geq c_0 - \sum_{t<i} d_t \geq d_i$. ($\Rightarrow$) Suppose $c_0(j) < \sum_{t \leq N} d_t(j)$ and let i^* be the first i with $\sum_{t \leq i} d_t(j) > c_0(j)$. Let A be the event that no bomb before round i^* sends anything to j. By the chain rule and diffuseness, $\mathbb{P}(A) \geq \eta^{i^*-1} > 0$. On A, either an earlier round stalls, or $c_{i^*-1}(j) = c_0(j) - \sum_{t<i^*} d_t(j) < d_{i^*}(j)$ and round i^* stalls. □

Under D0 the expected 20-round demand on +1 alone is $20 \cdot \frac{177}{202} \approx 17.5$, against $c_0(+1) = 5$, so a sure guarantee is out of the question with random rewards. The best one can hope for is a probabilistic guarantee, and its sign is decided brick by brick by the *drift*

$$\mu_j := \mathbb{E}r(j) - \mathbb{E}d(j) \qquad \text{(per round, } r(j) := 0 \text{ when } \xi \leq 0\text{)}.$$

Theorem 5.4 (Negative drift: exponential collapse)**.** *Let the rounds be i.i.d. and let rewards depend only on ξ and fresh randomness. If $\mu_j < 0$ for some j, then for $N > c_0(j)/|\mu_j|$*

$$\mathbb{P}(\textit{strong survival over } N) \leq \exp\Big(-\frac{2\,(N|\mu_j| - c_0(j))^2}{N\,s_j^2}\Big),$$

where s_j is the range of $r(j) - d(j)$.

Proof. Let $X_t = r_t(j) - d_t(j)$ (the reward drawn as if round t were solved). The X_t are i.i.d. with mean μ_j. On the survival event (3) gives $0 \leq c_N(j) = c_0(j) + \sum_{t \leq N} X_t$, so $\sum_t (X_t - \mu_j) \geq N|\mu_j| - c_0(j) > 0$. Hoeffding's inequality bounds the probability of this. □

For D0 with the unbiased split, $\mu_{\pm\times} = \frac{7}{16} - \frac{57}{101} = -\frac{205}{1616} \approx -0.127$ and $\mu_{\pm 1} = \frac{7}{16} - \frac{177}{202} = -\frac{709}{1616} \approx -0.439$: every brick drifts down, and the constant bricks drift fastest. Hoeffding is loose at $N = 20$; simulation (Table 2) shows that only 0.17% of games are survived and that the first stall comes at round 5.5 on average. When all drifts are positive, the classical Cramér–Lundberg argument gives a bound that holds for an unlimited number of rounds:

Theorem 5.5 (Positive drift: Lundberg bound)**.** *Under the assumptions of Theorem 5.4, let $\theta_j > 0$ satisfy $\mathbb{E}[e^{-\theta_j r(j)}]\,\mathbb{E}[e^{\theta_j d(j)}] \leq 1$. Then the probability that brick j ever causes a stall is at most $e^{-\theta_j(c_0(j) - d_j^{\max}+1)}$, and $\mathbb{P}(\textit{some stall ever}) \leq \sum_j e^{-\theta_j(c_0(j)-d_j^{\max}+1)}$.*

Proof. Let $W_i = c_0(j) + \sum_{t<i} r_t(j) - \sum_{t \leq i} d_t(j)$. If the first stall happens in round i because of j, then all earlier rounds were solved and $W_i = c_{i-1}(j) - d_i(j) < 0$. Write $W_i = W_1 + \sum_{t=2}^{i} Z_t$ with $Z_t = r_{t-1}(j) - d_t(j)$. Different Z_t involve disjoint rounds, and within a round ξ and (k_b, k) are drawn independently, so the Z_t are i.i.d. with $\mathbb{E}e^{-\theta_j Z} \leq 1$. Hence $M_i = e^{-\theta_j(W_i - W_1)}$ is a non-negative supermartingale. Let $\tau = \inf\{i : W_i < 0\}$; since W is integer-valued, $M_\tau \geq e^{\theta_j(W_1+1)}$ on $\{\tau < \infty\}$. Optional stopping gives $\mathbb{P}(\tau \leq n)\, e^{\theta_j(W_1+1)} \leq \mathbb{E}M_{\tau \wedge n} \leq 1$. Let $n \to \infty$ and use $W_1 \geq c_0(j) - d_j^{\max}$. The last claim is a union bound. □

Example 5.6. For the rebalanced generator D1 with the random split, all drifts are positive ($\frac{19}{144}$ for $\pm\mathsf{x}$ and $\frac{11}{144}$ for ± 1). The adjustment coefficients are $\theta_{\pm\mathsf{x}} = 0.127$ and $\theta_{\pm 1} = 0.066$. To push the Lundberg bound below 0.35, a stock of 30 *per brick* is needed. Probabilistic guarantees with random rewards are expensive, because the buffer grows like $\theta^{-1}\log(K/\varepsilon)$.

Remark 5.7 (Matching principle). If a rule sends a share π_j of each reward to brick j, then $\mu_j = \pi_j \mathbb{E}\xi^+ - \mathbb{E}d(j)$. The uniform shares $\pi_j = \frac{1}{4}$ ignore that, in D0, the constant bricks are needed 1.55 times as often as the x-bricks. Making every $\mu_j \geq 0$ then needs $\mathbb{E}\xi^+ \geq 4\max_j \mathbb{E}d(j) = 3.50$, whereas the balance law only asks for 2.88. Demand-proportional shares $\pi_j = \mathbb{E}d(j)/\mathbb{E}\delta$ give $\mu_j = \pi_j(\mathbb{E}\xi^+ - \mathbb{E}\delta)$, so every brick inherits the sign of the global balance.

Remark 5.8 (Implementation detail). The game shuffles the bricks by sorting them with a random comparator,

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sort(() => Math.random() - .5),
```

which is not a uniform shuffle. Measured in V8 (Node 22, the engine of Chrome), a one-part reward goes to $+\mathsf{x}$, $-\mathsf{x}$, $+1$, -1 with probabilities 0.36, 0.14, 0.19, 0.31. This lowers $\mu_{-\mathsf{x}}$ from -0.127 to about -0.160. In the test game $-\mathsf{x}$ was the first brick to run out. A Fisher–Yates shuffle removes the bias.

6 Designs with sure guarantees

Theorem 5.3 shows that a sure guarantee needs either a reward rule that knows what is needed, or a generator that knows what is in stock. We treat both.

6.1 Deficit filling and the Lindley queue

Definition 6.1. *Deficit filling* sends each reward unit, one at a time, to a brick minimising $c(j) - c_0(j)$, the count relative to its starting level (ties broken arbitrarily). The *total deficit* is $T(c) = \sum_j (c_0(j) - c(j))^+$.

Lemma 6.2. *Under deficit filling, if round i is solved then $T(c_i) \leq \big(T(c_{i-1}) + \delta_i - {\xi_i}^+\big)^+$.*

Proof. Paying d_i raises each term $(c_0(j) - c(j))^+$ by at most $d_i(j)$, so T grows by at most δ_i. Now add the ${\xi_i}^+$ reward units one by one. While $T > 0$, some brick is below its starting level, so the chosen brick is too, and the unit lowers T by exactly one. When $T = 0$ it stays 0. □

Theorem 6.3 (Main theorem: sure survival under deficit filling)**.** *Define the Lindley sequence $\Lambda_0 = 0$, $\Lambda_i = (\Lambda_{i-1} + \delta_i - {\xi_i}^+)^+$. Under deficit filling the perfect player strongly survives rounds $1, \ldots, N$ whenever*

$$\Lambda_{i-1} \;\leq\; \min_j (c_0(j) - d_i(j)) \qquad (1 \leq i \leq N). \tag{6}$$

In particular it suffices that $\min_j c_0(j) \geq L + \max_{i<N} \Lambda_i$, where $L = \max_{i,j} d_i(j)$.

Proof. By induction, assume rounds $1, \ldots, i-1$ were solved. Since $z \mapsto (z+a)^+$ is monotone, Lemma 6.2 gives $T(c_{i-1}) \leq \Lambda_{i-1}$. Then for every j, $c_{i-1}(j) \geq c_0(j) - (c_0(j) - c_{i-1}(j))^+ \geq c_0(j) - T(c_{i-1}) \geq c_0(j) - \Lambda_{i-1} \geq d_i(j)$ by (6), so round i is solvable. □

Equation (6) is the requested constraint on the solutions: it couples each ξ_i to the demands only through the scalar queue Λ. A negative ξ_i is a *debt* δ_i joining the queue, and a positive ξ_i with $\xi_i > \delta_i$ *serves* it. The simplest sequences keep the queue short by placing each debt directly before its repayment:

Corollary 6.4 (Debt, then repayment)**.** *Suppose the rounds form consecutive blocks of two kinds:*

- *a single round with $\xi \geq \delta$;*
- *a pair (negative round, positive round) whose positive solution repays both, $\xi_{\text{pos}} \geq \delta_{\text{pos}} + \delta_{\text{neg}}$.*

Then $\Lambda_i \leq P := \max \delta$ for all i, so $c_0(j) \geq L + P$ guarantees strong survival under deficit filling. With $P = 3$ and $|k_b|, |k| \leq 2$ (so $L = 2$), $c_0 = (5, 5, 5, 5)$ is enough. All solutions stay in $\pm\{1, \ldots, 6\}$, because a repayment needs at most $\delta_{\text{pos}} + \delta_{\text{neg}} \leq 6$, and strict alternation makes exactly half of the solutions negative.

Proof. Single blocks keep $\Lambda = 0$. In a pair, $\Lambda = \delta_{\text{neg}} \leq P$ after the negative round and $\Lambda = (\delta_{\text{neg}} + \delta_{\text{pos}} - \xi_{\text{pos}})^+ = 0$ after the positive one. Condition (6) reads $0 \leq c_0(j) - d_i(j)$ in the first round of a block and $\delta_{\text{neg}} \leq c_0(j) - d_i(j)$ in the second, both implied by $c_0(j) \geq L + P$. □

Remark 6.5 (Repay, do not prepay). Before any debt exists, deficit filling spreads surplus evenly over the K bricks, so raising every brick by D costs KD units. Repaying an existing debt of D costs D. Debts should therefore come *before* their repayments. This is convenient: the generator, which builds equations backwards, already knows the debt when it picks the next positive solution.

Proposition 6.6 (Random sequences under deficit filling)**.** *If the rounds are i.i.d. with $\mathbb{E}\xi^+ > \mathbb{E}\delta$, then Λ is a stable queue and $\mathbb{P}(\Lambda_i \geq b) \leq e^{-\theta b}$ for all i, where $\theta > 0$ solves $\mathbb{E}e^{\theta(\delta - \xi^+)} = 1$. Hence $\mathbb{P}(a \text{ stall within } N \text{ rounds}) \leq N\, e^{-\theta(\min_j c_0(j) - L + 1)}$.*

Proof. By the Lindley representation, Λ_i has the law of $\max_{0 \leq k \leq i}$ of the partial sums of the reversed increments, which is dominated by the supremum of a random walk with negative drift. The Lundberg argument of Theorem 5.5 bounds that supremum. A stall in round i needs $\Lambda_{i-1} \geq \min_j c_0(j) - L + 1$; take a union bound over $i \leq N$. □

For D1, $\theta = 0.092$; the bound is loose, but simulation shows a large real effect (Table 2). With $c_0 = (4, 4, 4, 4)$, deficit filling raises the share of survived 20-round games from 16% to 66%, and with $c_0 = (6, 6, 6, 6)$ to 89%.

6.2 Adaptive generation

Definition 6.7. Fix $a \geq 1$. The *adaptive generator* G_a draws round i from a base generator conditioned on the admissible set

$$\mathcal{A}(c_{i-1}) = \{(p, q, \xi) : \; d(p, q) \leq c_{i-1}, \quad \xi^+ \geq \delta(p, q) + a - \|c_{i-1}\|\}.$$

Theorem 6.8 (Sure survival under any allocation)**.** *Suppose the base generator gives positive probability to every cost-one pair and to some positive solution (independently of the pair), and that $\|c_0\| \geq a$. Then G_a is well defined in every round, and the perfect player strongly survives every number of rounds, under* every *allocation rule, adversarial ones included.*

Proof. By induction, assume $\|c_{i-1}\| \geq a \geq 1$. Some brick j has $c_{i-1}(j) \geq 1$, and the cost-one pair using only j, together with any positive ξ of the base law, satisfies $\xi^+ \geq 1 \geq 1 + a - \|c_{i-1}\|$. So $\mathcal{A}(c_{i-1})$ has positive mass and round i is drawn from it. Its equation is affordable because $d_i \leq c_{i-1}$. Whatever the split, $\|r_i\| = {\xi_i}^+$, so $\|c_i\| = \|c_{i-1}\| - \delta_i + {\xi_i}^+ \geq a$. □

The guarantee is unconditional, but by Theorem 4.2 the generator can only buy it by skewing the sequence: under a deficit it must offer more positive solutions or cheaper equations. Simulation measures this skew (Table 2). On D0, G_1 lowers the share of negative solutions from 0.50 to 0.47 (20 rounds) and 0.44 (100 rounds), and the mean demand from 2.88 to 2.34 and 2.09. On the balanced D1, G_4 is almost invisible: $q = 0.49$–0.50, $\bar{\delta} = 2.25$–2.27. The adaptive generator is best used as a safety net under a generator that already satisfies the balance law.

7 Trash, Game Over and the value of a negative bomb

Let the well have R rows, with Game Over when a trash block lands in one of the top two rows (above the red line). Write $H = R - 2$.

Lemma 7.1 (Height)**.** *Each trash landing raises the stack height by at most one, while clearing and settling never raise it. Game Over therefore needs at least $H + 1$ trash blocks.*

Proof. A block lands one row above the highest column it covers. So if the stack height is h, it lands in row $r \geq R - h - 1$ and the new height is at most $h + 1$. Clearing removes bricks and settling moves them down. Game Over means $r < 2$, which requires $h \geq R - 2 = H$ before the landing, i.e. at least H earlier landings. □

The game sizes the well to the screen. Measured values of H are 18 (390×844), 19 (412×915), 16 (360×740), 13 (375×667) and 12 (1366×860). Equation blocks are $3, 4, 5$ cells wide with frequencies $0.38, 0.57, 0.05$, and on 9 columns they overlap heavily: the simulated mean number of trash blocks to Game Over is 23.6 for $H = 18$ and 16.1 for $H = 12$. Under D0 with $H = 18$, no simulated 20-round game reached Game Over, even counting no clearing at all. With $H = 12$ the rate is 5.5%.

Proposition 7.2 (Clearing depth)**.** *A negative bomb ξ clears the trash bricks whose centres lie within $\rho(\xi) = 0.8 + 0.7|\xi|$ cells of the bomb's centre. If it lands on a trash column at least $\lfloor \rho(\xi) \rfloor$ deep, that column loses its top $\lfloor \rho(\xi) \rfloor$ bricks, and in deep, flat trash it removes $\#\{(i, j) \in \mathbb{Z} \times \mathbb{Z}_{\geq 1} : i^2 + j^2 \leq \rho^2\}$ bricks:*

$\lvert\xi\rvert$	*1*	*2*	*3*	*4*	*5*	*6*
column depth	*1*	*2*	*2*	*3*	*4*	*4**
bricks in flat trash	*3*	*4*	*10*	*15*	*26*	*30*

**In floating point* $0.8 + 0.7 \cdot 6 = 4.999\ldots$*, so the depth is 4, not 5.*

Proof. The bomb's bottom rests on the column surface s and its centre is $\frac{1}{2}$ cell higher. A brick in the same column at row $s - 1 + j$ ($j \geq 1$) is at distance j, and one i columns aside at distance $\sqrt{i^2 + j^2}$. Count the lattice points. □

Proposition 7.3 (Paradox of the negative bomb)**.** *In any design that guarantees strong survival, the perfect player never produces trash, so their negative bombs never clear anything. Each negative round is then a pure cost of $\delta_i > 0$ units.*

The proof is immediate. The proposition says that clearing only has value for imperfect play: time pressure, wasted operations, slow recognition. That is exactly where trash comes from in real games, so negative bombs act as the second chance for human mistakes, and their cost must be built into the balance law. Under the current rules a player can even *gain* by letting negative equations fall unsolved: each one costs a single trash block, and by Lemma 7.1 a phone with $H = 18$ tolerates 18 of them. The square-root bombs of Section 10 resolve this tension, because they refuel and clean at the same time.

8 Adding doubling (×2)

We now allow equations such as $\frac{1}{2}x = 5$ or $\frac{1}{2}x + 1 = 6$ and a brick $\times 2$ that multiplies both sides by 2. Write an equation as $U = \alpha x + \beta$ with $L = (\alpha - \Delta)x + (\beta + \Delta\xi)$, and let $u := \alpha - \Delta$ be the coefficient of x in L. The state is $s = (\alpha, \beta, \Delta) \in \mathbb{Q}^3$ with ξ fixed. The operations act as

$$+\mathsf{x}, -\mathsf{x}\colon \alpha \mapsto \alpha \pm 1, \qquad +1, -1\colon \beta \mapsto \beta \pm 1, \qquad \times 2\colon s \mapsto 2s, \qquad (\div 2\colon s \mapsto s/2 \text{ in Section 9}),$$

and, by the argument of Lemma 3.2, the equation is solved if and only if $s = (1, 0, 1)$.

Lemma 8.1 (Levels)**.** *In a sequence of operations, the* level ℓ_k *of a translation* k *is the number of doublings minus the number of halvings applied after it. If the sequence contains* σ *more doublings than halvings, its final state is*

$$\Big(2^\sigma\alpha + \sum_{k\in X} \varepsilon_k 2^{\ell_k},\ 2^\sigma\beta + \sum_{k\in C} \varepsilon_k 2^{\ell_k},\ 2^\sigma\Delta\Big),$$

where X *and* C *are the* x*- and constant translations and* $\varepsilon_k = \pm 1$ *their signs.*

Proof. All operations are affine maps $v \mapsto \lambda v + w$ on $\mathbb{Q}^3$. A translation applied at some moment is afterwards multiplied by 2 at every doubling and by $\frac{1}{2}$ at every halving; induct over the sequence. □

Definition 8.2. For $m \in \mathbb{N}$ and $Z \in \mathbb{Z}$, the *m-capped signed-binary weight* $w_m(Z)$ is the least number of terms $\pm 2^\ell$ with $0 \le \ell \le m$ that sum to Z.

Lemma 8.3 (Properties of w_m)**.**

(i) $w_0(Z) = |Z|$*. For* $m \ge 1$*,* $w_m(Z) = w_{m-1}(Z/2)$ *if* Z *is even, and* $w_m(Z) = 1 + \min\{w_{m-1}(\frac{Z-1}{2}),\ w_{m-1}(\frac{Z+1}{2})\}$ *if* Z *is odd.*

(ii) $\lceil |Z|/2^m \rceil \le w_m(Z) \le \lfloor |Z|/2^m \rfloor + m$*.*

(iii) $w_m(2^m Z) = |Z|$*, and only terms* $\pm 2^m$ *of one sign attain it.*

Proof. (i) Take an optimal representation. Two level-0 terms of opposite signs cancel and can be deleted. Two of the same sign can be replaced by one level-1 term. So an optimum has at most one level-0 term, and the parity of Z decides whether it has one. Removing it and halving the rest gives a representation of $(Z \mp 1)/2$ or $Z/2$ with levels in $[0, m-1]$, and conversely. (ii) Each term is at most 2^m in absolute value. For the upper bound, use $\lfloor |Z|/2^m \rfloor$ terms $\pm 2^m$ and the binary digits of the remainder, which is below 2^m. (iii) The lower bound in (ii) is $|Z|$, and it is attained only if every term has absolute value 2^m and the same sign. □

Theorem 8.4 (Cost with doubling)**.** *With* $J' = \{+\mathsf{x}, -\mathsf{x}, +1, -1, \times 2\}$*, an equation is solvable if and only if* $\Delta = 2^{-m}$ *for some* $m \in \mathbb{N}$ *and* $A := 2^m u$*,* $B := 2^m \beta$ *are integers. Its cheapest cost is*

$$C(E) = m + w_m(A) + w_m(B).$$

Proof. Being solved requires $2^\sigma\Delta = 1$ by Lemma 8.1, and $\sigma \ge 0$ is the number of doublings, so $\Delta = 2^{-m}$ with exactly m doublings and all levels in $[0, m]$. Since $2^m\Delta = 1$, the solved condition $\alpha_{\text{final}} = 1$ reads $\sum_X \varepsilon_k 2^{\ell_k} = 1 - 2^m\alpha = -A$, and $\beta_{\text{final}} = 0$ reads $\sum_C \varepsilon_k 2^{\ell_k} = -B$. Both sums are integers, which gives necessity, and every solving sequence costs at least $m + w_m(A) + w_m(B)$ because w_m is even in Z. Conversely, take optimal representations of $-A$ and $-B$. For $\ell = m, m-1, \ldots, 0$, perform the translations of level ℓ and then (if $\ell > 0$) one doubling. This realises the bound. □

Both formulas, this one and the halving version of Theorem 9.1, agree with breadth-first search over all 17 474 states within 9 operations of the solved state, with no exceptions (Section A).

Proposition 8.5 (Consequences of doubling)**.**

(a) Order matters. *If* $u, \beta \in \mathbb{Z}$ *(integer constants), then* $C(E) = m + |u| + |\beta|$*, attained only by doing all translations* before *any doubling. The textbook move of doubling first to clear the fraction costs* $m + 2^m(|u| + |\beta|)$*, which is* $(2^m - 1)(|u| + |\beta|)$ *extra units. For example* $\frac{1}{2}x + 1 = 6$ *takes 2 operations as* $-1, \times 2$ *but 3 as* $\times 2, -1, -1$*.*

(b) Larger bombs. *If all displayed constants are integers, then* $\xi \in 2^m\mathbb{Z}$*, since* $\Delta\xi$ *is a difference of integer constants. A doubling equation therefore pays* $|\xi| \ge 2^m$*: with solutions in* $\pm\{1, \ldots, 6\}$ *and* $m = 1$*, only* $\pm 2, \pm 4, \pm 6$ *occur, with mean positive reward* 4*.*

(c) Separate budget. *The* $\times 2$ *brick is consumed exactly* m *times per equation, so it needs its own balance:* $\mathbb{E}r(\times 2) \ge \mathbb{E}m$*.*

Proof. (a) By Lemma 8.3(iii) with $A = 2^m u$ and $B = 2^m \beta$, only level-m terms are optimal, i.e. translations made before the first doubling. Doubling first puts all translations at level 0, where each is worth 1. (b) The constant of L is $\beta + \Delta\xi$; if it and β are integers, so is $2^{-m}\xi$. (c) Every solving sequence contains exactly m doublings. □

9 Adding halving (÷2)

Halving allows integer equations with $\Delta = 2^m$, such as $2x+4 = 10$, $3x+1 = x+7$ or $4x+3 = 11$.

Theorem 9.1 (Cost with halving)**.** *With $J'' = \{+\mathsf{x}, -\mathsf{x}, +1, -1, \div 2\}$, an equation is solvable if and only if $\Delta = 2^m$ for some $m \in \mathbb{N}$ and $u, \beta \in \mathbb{Z}$. Its cheapest cost is $C(E) = m + w_m(u) + w_m(\beta)$.*

Proof. Now $\sigma = -m$ and the levels lie in $[-m, 0]$. The solved condition reads $\sum_X \varepsilon_k 2^{\ell_k} = 1 - 2^{-m}\alpha = -2^{-m}u$. Multiplying by 2^m gives $\sum_X \varepsilon_k 2^{\ell_k + m} = -u$ with exponents in $[0, m]$, and likewise for β. The rest is as in Theorem 8.4. In words, a translation made after h halvings is worth 2^h in the original equation. □

Example 9.2. $2x + 4 = 10$: $1 + w_1(4) = 3$ ($\div 2, -1, -1$; subtracting first takes 5). $2x + 3 = 7$: $1 + w_1(3) = 3$ ($-1, \div 2, -1$: make the constant even first). $3x + 1 = x + 7$: $1 + w_1(1) + w_1(1) = 3$ ($-\mathsf{x}, -1, \div 2$). $4x + 3 = 11$: $2 + w_2(3) = 4$, with *two incomparable* cheapest demands: $-1, \div 2, -1, \div 2$ uses (-1:2, $\div$2:2), and $+1, \div 2, \div 2, -1$ uses ($+1$:1, -1:1, $\div$2:2).

Proposition 9.3 (Economics of halving)**.**

(a) Halving cuts the constant demand from $|\beta|$ to $w_m(\beta) \in [\lceil |\beta|/2^m \rceil, \lfloor |\beta|/2^m \rfloor + m]$, at the price of m units of $\div 2$, and it puts no restriction on ξ. Large constants become cheap.

(b) The set $\mathcal{D}(E)$ of demand vectors of solving sequences can have several incomparable minimal elements (Example 9.2). The equation is affordable if and only if $c \in \mathcal{D}(E) + \mathbb{N}^{J''}$, so Corollary 3.4 fails: the perfect player has to choose *which bricks to spend. In Example 9.2 a stock with one -1 and one $+1$ can only use the second solution.*

Proof. (a) is Lemma 8.3(ii) together with ξ being free in L's constant $\beta + 2^m\xi$. (b) is the example. □

Proposition 9.4 (Both scalings: detours)**.** *With both $\times 2$ and $\div 2$ available, an equation with $\Delta = 1$ might be solved more cheaply by a detour through other scales. The cheapest cost is still $|u| + |\beta|$ whenever $|u| + |\beta| \le 5$, and the bound is sharp: $x + 6 = \xi + 6$ takes 5 operations ($\div 2$, three -1, $\times 2$) instead of 6.*

Proof. A sequence with scalings has $h \ge 1$ doublings and h halvings, and by Lemma 8.1 its translation terms are at most 2^h in absolute value. Its cost is therefore at least $g(h) = 2h + \lceil |u|/2^h \rceil + \lceil |\beta|/2^h \rceil$. Write $S = |u| + |\beta| \le 5$. For $h = 1$, $g \ge 2 + \lceil S/2 \rceil \ge S$ (check $S = 5$: $2 + 3 = 5$). For $h \ge 2$, $g \ge 4 + [S > 0] \ge S$. The example is verified directly, and the search in Section A finds no shorter detour for $S \le 5$. □

10 Adding square roots: ±-bombs

Consider equations $(x - s)^2 + k = t^2 + k$ with $s, k \in \mathbb{Z}$ and $t \in \mathbb{Z}_{>0}$, for example $x^2 = 4$, $x^2 + 1 = 10$ or $(x - 1)^2 = 9$. A brick $\surd$ applies only to an equation of the form $(x - s)^2 = t^2$ and turns it into $x - s = \pm t$, which the game carries on as one ±-equation: both branches receive the same operations.

Lemma 10.1 (Cost of a quadratic)**.** *With the bricks $+\mathsf{x}, -\mathsf{x}, +1, -1, \surd$, the cheapest cost is $|k| + 1 + |s|$. It uses $|k|$ units of -1 if $k > 0$ (of $+1$ if $k < 0$) before the root, one $\surd$, and $|s|$ units of $+1$ if $s > 0$ (of -1 if $s < 0$) after it. Solved, it gives the two solutions $\xi_{1,2} = s \pm t$.*

Proof. Before $\surd$ only translations act. To reach the form "square = constant", the x-translations must cancel, since otherwise the right side contains x, and the constant translations must total $-k$. That takes at least $|k|$ operations, and exactly $|k|$ without x-operations. After $\surd$ both branches are level-one equations with $(p, q) = (0, -s)$, so Proposition 3.3 gives $|s|$ further operations. □

A solved $\pm$-equation produces a $\pm$-*bomb*, best shown as two bombs falling side by side. It returns $\varrho = (s + t)^+ + (s - t)^+$ units and clears with each negative root.

Proposition 10.2 (Self-financing quadratics)**.** *A quadratic pays for itself ($\varrho \geq |k| + 1 + |s|$) if and only if:*

- *for $|s| < t$ (roots of opposite signs): $t \geq |k| + 1 + |s| - s$, i.e. $t \geq |k| + 1$ when $s \geq 0$ and $t \geq |k| + 1 + 2|s|$ when $s < 0$;*
- *for $s > t$ (both roots positive): $s \geq |k| + 1$;*
- *for $s < -t$ (both roots negative): never.*

($s = \pm t$ gives the root 0, which the generator excludes.) In particular a symmetric quadratic ($s = 0$) with $t \geq |k| + 1$ both refuels and produces the negative bomb $-t$.

Proof. For $|s| < t$, $\varrho = s + t$. For $s > t$, $\varrho = 2s$ and $|s| = s$. For $s < -t$, $\varrho = 0$. □

Theorem 10.3 (Quadratics as stabilisers)**.** *Mix the linear rounds of a generator with drift $\mu_{\text{lin}} = \mathbb{E}[\xi^+ - \delta] < 0$ and a fraction φ of symmetric quadratics with drift $\mu_{\text{q}} = \mathbb{E}t - 1 - \mathbb{E}|k| > 0$. The mixture satisfies the balance law in expectation if and only if*

$$\varphi \;\geq\; \frac{|\mu_{\text{lin}}|}{|\mu_{\text{lin}}| + \mu_{\text{q}}}.$$

For D0 ($\mu_{\text{lin}} = -\frac{457}{404}$) with t uniform on $\{2, \ldots, 6\}$ and k uniform on $\{-1, 0, 1\}$ ($\mu_{\text{q}} = \frac{7}{3}$), the threshold is $\varphi \geq 0.327$. Every quadratic round also yields one negative bomb, so the share of negative bombs stays at one half.

Proof. By linearity the mixed drift is $(1 - \varphi)\mu_{\text{lin}} + \varphi\mu_{\text{q}}$; solve for $\varphi \geq 0$. The numbers follow from Table 1. □

Square roots therefore resolve the paradox of Proposition 7.3: the negative half of a $\pm$-bomb is paid for by its positive half. The $\surd$ brick needs its own budget ($\mathbb{E}r(\surd) \geq \varphi$). The game must refuse $\surd$ on a non-square, such as $x^2 + 5 = 9$ before the 5 is removed, and that refusal is itself a lesson.

11 Recommendations

The constraints asked for in the introduction, collected:

C1. Balance (necessary, any rules). Every prefix must satisfy (4), and in the long run $(1 - q)\nu_+ \geq (1 + \varepsilon)\bar{\delta}$, with a margin ε for human waste and failures. Half the solutions can be negative only if the mean positive solution is about twice the mean number of operations. Example D1: at most 3 operations ($\bar{\delta} = \frac{7}{3}$), solutions in $\{-6, \ldots, -1\} \cup \{3, \ldots, 8\}$ ($\nu_+ = 5.5$), drift $+\frac{5}{12}$.

C2. Matching (with random rewards). Every brick needs $\mu_j \geq 0$. Either split rewards in proportion to demand (Remark 5.7) or use deficit filling. Use an unbiased shuffle in any case (Remark 5.8).

C3. Sure guarantee, option 1. Deficit filling with the Lindley condition (6). Concretely: every negative solution is followed by a positive one repaying both, $\delta \leq 3$, $|k_b|, |k| \leq 2$ and $c_0 = (5, 5, 5, 5)$. This keeps the original solutions $\pm\{1, \ldots, 6\}$ (Corollary 6.4).

Table 2: Monte Carlo results for a perfect player (20,000 runs for $N = 20$, 4,000 for $N = 100$, 5,000 for D2). "Survival" is the share of games without a stall. DF is deficit filling, G_a the adaptive generator and D2 the debt-then-repay design of Corollary 6.4. With the biased shuffle of the actual implementation, D0 survives 0.14% of 20-round games (200,000 runs).

		$N = 20$			$N = 100$		
design	c_0	survival	solved	1st stall	survival	solved	q, $\bar{\delta}$
D0 random split	3,3,5,5	0.002	9.6	5.5	0.000	18.3	0.50, 2.88
D0 + DF	3,3,5,5	0.035	11.3	7.5	0.000	17.2	0.50, 2.88
D0 + G_1	3,3,5,5	1	20	–	1	100	0.47, 2.34 / 0.44, 2.09
D1 random split	3,3,5,5	0.156	15.0	7.9	0.028	67.5	0.50, 2.33
D1 random split	4,4,4,4	0.162	15.2	8.4	0.031	68.1	0.50, 2.33
D1 + DF	4,4,4,4	0.656	18.4	9.2	0.518	86.7	0.50, 2.33
D1 + DF	6,6,6,6	0.885	19.7	12.3	0.760	95.6	0.50, 2.33
D1 + G_4	4,4,4,4	1	20	–	1	100	0.49, 2.25 / 0.50, 2.27
D2 + DF	5,5,5,5	1	20	–	1	100	0.50, ≤ 3

C4. Sure guarantee, option 2. Keep the random split for its fun, and let the generator see the stock (G_a, Theorem 6.8) on top of a generator that already satisfies C1. The distortion is then barely measurable.

C5. Give negative bombs a purpose. Under any sure guarantee they clear only the trash caused by human mistakes (Proposition 7.3). Accept this, or give negative solutions mostly through ±-bombs (Theorem 10.3).

C6. Extensions.

- $\times 2$ makes order matter and yields bombs that are multiples of 2^m.
- $\div 2$ makes large constants cheap and the cheapest demand non-unique, so the player has to choose which bricks to spend.
- $\sqrt{}$ gives self-financing ±-bombs; about a third of quadratics would already cancel the current deficit.

Each new brick adds its own budget line ($\mathbb{E}r(j) \geq \mathbb{E}d(j)$).

A Methods

Exact computations.

- The joint law of (k_b, k) after rejection, the demand moments of Table 1 and the reward laws of the split rule were computed with rational arithmetic. For D0, each pair with $|k_b| = 1$ and $|k| \geq 1$ has probability $\frac{5}{101}$; each pair with $|k_b| = 2$ (then $|k| \leq 2$), or with $k_b = 0$ and $|k| \in \{2, 3\}$, has $\frac{5}{202}$; the cost-one pairs have $\frac{1}{101}$ ($k_b = 0$) or $\frac{2}{101}$ ($k = 0$).
- $\bar{p}_N$ in Theorem 4.3 is an exact forward recursion over the total stock.
- The adjustment coefficients solve the stated moment equations by bisection.

Exhaustive search. Breadth-first search from the solved state $(1, 0, 1)$ over the inverse operations, with exact rationals, found 17 474 states within 9 operations, separately for $\{\pm\mathsf{x}, \pm 1, \times 2\}$ and for $\{\pm\mathsf{x}, \pm 1, \div 2\}$. On every state the distance equals the formula of Theorem 8.4 or Theorem 9.1. With both scalings, for $\Delta = 1$, the distance equals $|u| + |\beta|$ whenever this is at most 5, and the first detour gain appears at $|u| + |\beta| = 6$ (Proposition 9.4).

Simulation. The simulated player is perfect in the sense of Definition 2.1. An unsolvable equation becomes trash, and play continues with the next one.

- Rounds were generated exactly as in the game's code (D0) or as described for D1/D2.
- The split rule is the game's, with a Fisher–Yates shuffle except where the biased sort is stated.
- Trash stacking used the measured block widths (15,000 equations rendered in headless Chrome at five screen sizes) and uniformly random columns.
- The screen sizes and the shuffle bias were measured on the actual page and in Node 22 (V8).